Physics-Informed Neural Networks for Parameter Estimation and Hidden-State Recovery in an SEIRD Epidemic Model with Time-Varying Transmission

The 2014 West African Ebola Outbreak (Sierra Leone)

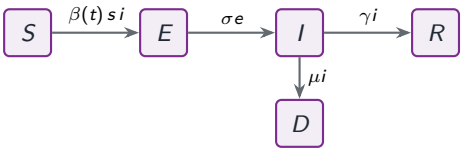
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Why transmission must change over time

The problem

- Ebola spreads by contact with body fluids, and with the dead at funerals
- isolation, safe burials and contact tracing cut transmission as the epidemic went on: β is **not** a constant
- surveillance reports only cases and deaths; S and E are never measured



$\dot{s} = -\beta s i, \quad \dot{e} = \beta s i - \sigma e, \quad \dot{i} = \sigma e - (\gamma + \mu) i, \quad \dot{r} = \gamma i, \quad \dot{d} = \mu i$

fractions of N, so $s + e + i + r + d = 1$ at all times

	Meaning	Value
N	population (2014 census)	7,092,113
σ	incubation rate (9.4 days)	0.1064 /day
γ	recovery rate (7 days)	0.1429 /day
μ	death rate	0.1000 /day
β	transmission rate (baseline)	0.4858 /day
CFR	$\mu/(\gamma + \mu)$	41.2%
R_0	$\beta/(\gamma + \mu)$	2.00

Assumptions

- closed population, homogeneous mixing
- recovered people stay immune
- σ, γ fixed from WHO clinical data, so the data only have to determine $\beta(t)$ and μ

Published models: how others handled $\beta(t)$

Module 1 ▪ each makes one assumption about how transmission changes, and each assumption has a cost

Chowell et al. (2004) Congo 1995, Uganda 2000

- SEIR model; β is high until control begins, then falls
- fitted to case counts: $R_0 \approx 1.83$ (Congo), 1.34 (Uganda)
- × the **shape** of the fall is chosen in advance

Legrand et al. (2007) three routes of infection

$$\lambda(t) = \beta_I \frac{I}{N} + \beta_H \frac{H}{N} + \beta_F \frac{F}{N}$$

- community (I), hospital (H) and funeral (F) transmission
- × H and F follow I, so from one national case curve the three β's trade off: **many sets fit equally well**

Rivers et al. (2014) a policy step

$$\beta(t) = \beta_0 \left[1 - \delta \, \mathbf{1}(t \geq t_{\text{policy}})\right]$$

- β drops by a fraction δ on a chosen date ($\mathbf{1}$ switches from 0 to 1 there)
- × an **instant jump** at an **assumed date**; real behaviour changes over weeks

Althaus (2014) exponential decay

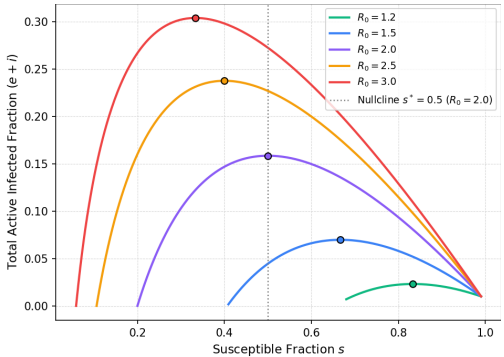
$$\beta(t) = \beta_0 \, e^{-k(t-t_0)}, \qquad \frac{dR_t}{dt} = -\frac{\beta s}{\gamma + \mu}\left[k + \beta i\right] < 0$$

- smooth, but R_t can **only fall**
- × once $R_t < 1$ it never rises again: **no resurgence**, no relaxation of control

The gap: every published model fixes the *shape* of $\beta(t)$ in advance (a date, a step, a decay). Our PINN learns $\beta(t)$ from cases and deaths, with no date or shape assumed.

Phase plane: where the epidemic peaks

Module 1 ▪ exact formulas, checked against an accurate ODE solver (DOP853)



$R_0 = 1.2$–3.0; time runs right to left; dots = predicted peaks

Divide the equations: i cancels exactly

$$\frac{d(e+i)}{ds} = -1 + \frac{1}{R_0\, s}$$

Peak where the slope is zero: $s^* = 1/R_0$

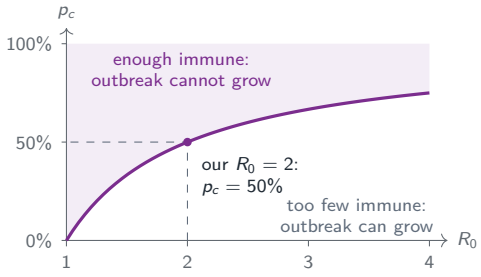
$$(e+i)_{\max} = (e_0 + i_0) + s_0 - \tfrac{1}{R_0}\left[1 + \ln(R_0 s_0)\right]$$

- $R_0 = 2$: peak **15.85%** at $s = 0.5$ (i alone about 4.8%)
- formula vs. ODE: relative error at most 1.3×10^{-5}
- right of s^*: $e + i$ grows ($R_t > 1$); left of s^*: it shrinks ($R_t < 1$)

Why it matters: the peak comes exactly when the susceptible share falls to $1/R_0$, the herd-immunity threshold (next slide). These exact formulas are also the checks used for the solvers (Module 3) and the PINN (Module 4).

Herd immunity and the control threshold

Module 1 ▪ how many must be immune, or how much transmission must be cut, before an outbreak cannot grow



R_0	1.5	2	3	4
$p_c = 1 - 1/R_0$	33%	**50%**	67%	75%

Where it comes from

- one case infects $R_t = R_0\, s$ people (only susceptible contacts count), so growth stops once $s < 1/R_0$: the immune share must exceed $p_c = 1 - 1/R_0$
- same threshold as Module 2: a disease-free state is stable only if $R_0 s^* < 1$

Two ways to reach it

- **immunity**, by vaccination or past infection; there was no licensed Ebola vaccine in 2014–15
- **cutting transmission** by δ: $R_t = (1-\delta)R_0\, s < 1$ when $\delta > \delta_c = 1 - 1/R_0$ (50% for $R_0 = 2$)

Two subtleties

- **overshoot:** reaching p_c only marks the *peak*; those still infectious keep infecting, so with $R_0 = 2$ about 80% are infected in the end, not 50%
- **timing:** later on less is needed, $1 - 1/R_t$: 37.5% at $s = 0.8$

What it means for control: without a vaccine, isolation, safe burials and tracing together must cut transmission by **more than 50%** at the start, and the earlier they act, the smaller the epidemic.

R_0 and stability: will an outbreak grow?

Next-generation matrix: split the (e, i) equations into new infections F and moves between classes V

$$F = \begin{pmatrix} 0 & \beta s^* \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} \sigma & 0 \\ -\sigma & \gamma + \mu \end{pmatrix}$$

$$R_0 = \rho(FV^{-1}) \text{ at } s^* = 1 \;=\; \frac{\beta}{\gamma + \mu}$$

- rate × time: 0.486 /day × 4.1 infectious days ≈ **2.00**
- σ drops out: incubation changes *when* people infect others, not *how many*
- over time: $R_t = \beta(t)\, s(t)/(\gamma + \mu)$, which the PINN gives us

Early growth rate = the positive eigenvalue

$\lambda_+ = 0.062734$ /day (theory), 0.062732 (simulation)

- doubling time **11.1 days**; $R_0 = (1 + \frac{\lambda_+}{\sigma})(1 + \frac{\lambda_+}{\gamma+\mu})$ gives back 2.0004

Equilibria: every one is disease-free, $(s^*, 0, 0, r^*, d^*)$

no births and lasting immunity, so there is no endemic state

s^*	$R_0 s^*$	eigenvalues (e, i block)	result
1.00	2.00	-0.412, $+\mathbf{0.063}$	unstable
0.75	1.50	-0.383, $+\mathbf{0.034}$	unstable
0.40	0.80	-0.334, -0.016	stable
0.25	0.50	-0.307, -0.042	stable

- threshold $R_0 s^* = 1$: herd immunity once $s^* < 1/R_0 = 50\%$
- **stable, not asymptotically**: three eigenvalues are 0, so the system settles at a *nearby* disease-free state

One number decides growth: a few infections grow if $R_0 s^* > 1$ and die out if $R_0 s^* < 1$. The growth rate in early case data gives R_0 back, and our simulation matches the theory to 3×10^{-5}.

Final size: how many people get infected in total?

Module 2 ▪ an exact answer without simulating, and a test for every numerical method in the project

Divide $\dot{s}$ by $\dot{r}+\dot{d}$: i cancels

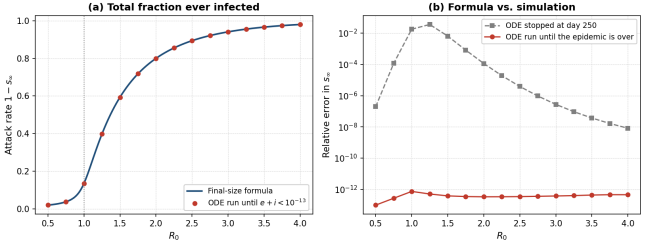
$$\frac{ds}{d(r+d)} = \frac{-\beta s i}{(\gamma+\mu)\, i} = -R_0\, s$$

Integrate to the end of the epidemic

$$\ln \frac{s_0}{s_\infty} = R_0\,(1 - s_\infty - r_0 - d_0)$$

$$d_\infty = d_0 + \underbrace{\frac{\mu}{\gamma+\mu}}_{\text{CFR } 41\%}\,(1 - s_\infty - r_0 - d_0)$$

- $R_0 = 2$: **80%** infected by the end, **33%** die (no interventions)
- exactly one root in $(0, s_0)$; found with Brent's method



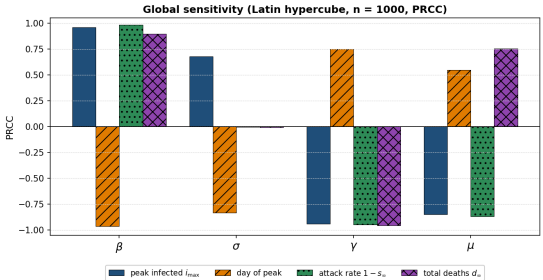
(a) formula (line) and ODE (dots); (b) error: run to the end (red) vs. stop at day 250 (grey)

- ✓ run the ODE until $e + i < 10^{-13}$: formula and ODE agree to $\mathbf{10^{-12}}$ for R_0 from 0.8 to 3.5
- × stop at day 250: errors up to **3.6%** near $R_0 = 1.25$, because slow epidemics last up to ∼1000 days

The catch: the β cancels only if it is constant. With $\beta(t)$ there is no closed formula, which is one reason we need a data-driven method such as the PINN.

Sensitivity: which parameters matter most?

Module 2 ▪ Latin hypercube sampling ($n = 1000$, seed 42) and partial rank correlation (PRCC)



PRCC = one parameter's effect with the others held fixed

	peak	peak day	attack rate	deaths
β	$+0.96$	-0.96	$+0.98$	$+0.90$
σ	$+0.68$	-0.83	$0.00^{\text{n.s.}}$	$-0.01^{\text{n.s.}}$
γ	-0.94	$+0.75$	-0.95	-0.95
μ	-0.85	$+0.55$	-0.87	$+0.75$

n.s. not significant at the 1% level.

Local index of R_0 (% change per 1%): β +1, γ −0.59, μ −0.41, σ 0.

- β raises everything: the main lever for control
- γ, μ shrink the epidemic, but μ **raises** deaths
- σ moves the peak but not the final size, **as the formula predicts**

What this means for the PINN: cases and deaths alone cannot separate σ, γ and μ. So we fix σ and γ from clinical data; the PINN learns $\beta(t)$, and μ from the share of cases that die.

Two solvers written from scratch, one trusted reference

Forward Euler one slope per step

$\mathbf{y}_{k+1} = \mathbf{y}_k + h\, f(t_k, \mathbf{y}_k)$; local error $O(h^2)$, global error $O(h)$

Classical Runge–Kutta (RK4) four slopes per step

$$k_1 = f(t_k, \mathbf{y}_k), \qquad k_2 = f\left(t_k + \tfrac{h}{2},\ \mathbf{y}_k + \tfrac{h}{2} k_1\right),$$
$$k_3 = f\left(t_k + \tfrac{h}{2},\ \mathbf{y}_k + \tfrac{h}{2} k_2\right), \qquad k_4 = f(t_k + h,\ \mathbf{y}_k + h\, k_3)$$
$$\mathbf{y}_{k+1} = \mathbf{y}_k + \tfrac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

- local error $O(h^5)$, global error $O(h^4)$
- both update all five compartments at once with NumPy arrays

Reference solution

- adaptive 8th-order Dormand–Prince (**DOP853**, SciPy)
- rtol 3×10^{-14}, atol 10^{-16}: close to the smallest SciPy accepts

Why so tight?

- a reference must be far more accurate than the solvers it judges
- with rtol $= 10^{-12}$, RK4's error at the smallest step reaches the reference's own ($\approx 10^{-12}$) and its measured order falls to **3.74**
- the fault is in the reference, not in RK4

Idea: Euler is the simplest method (order 1), RK4 the accurate workhorse (order 4). Both orders are known in advance, so the measured slopes test the code.

Two checks: conservation, and how accuracy is measured

Conservation of the total

$$\frac{d}{dt}(s + e + i + r + d) = 0$$

- the components of f sum to zero, so every Euler or RK4 step keeps $s + e + i + r + d = 1$, up to rounding

Result $t \in [0, 60]$, $h \in \{1,\ 0.25,\ 0.03125\}$

$$\max_k \left| \sum_j y_{k,j} - 1 \right| \ \leq\ 1.2 \times 10^{-15}$$

- a few multiples of machine epsilon (2.2×10^{-16})
- shows f is coded consistently; it does **not** measure accuracy

Experiment E1

- $\mathbf{y}_0 = (0.99,\ 0.005,\ 0.005,\ 0,\ 0)$ over $t \in [0, 60]$ days
- $h = 1,\ 0.5,\ 0.25,\ 0.125,\ 0.0625,\ 0.03125$ days

Global error: largest miss over all times and all five compartments

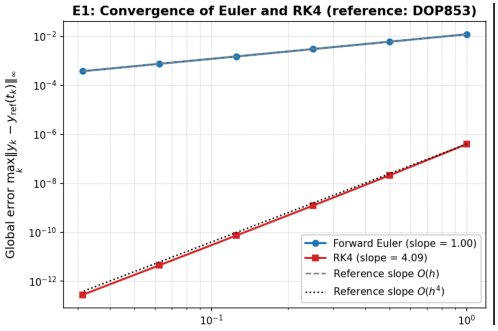
$$E(h) = \max_k \max_j \left| y_{k,j} - y_{\mathrm{ref},j}(t_k) \right|$$

Observed order: if $E(h) \approx Ch^p$, halving h gives $p \approx \log_2\left[E(h)/E(h/2)\right]$; a least-squares fit of $\log E$ on $\log h$ gives one overall slope.

Two separate questions: conservation tests that the equations are coded consistently; convergence tests that the error shrinks at the theoretical rate.

E1 result: Euler converges at order 1, RK4 at order 4

Module 3 ▪ global error against the DOP853 reference, log–log axes



reference lines: slopes 1 and 4

h (days)	Euler error	order	RK4 error	order
1	1.22×10^{-2}	1.004	4.00×10^{-7}	4.225
0.5	6.09×10^{-3}	1.002	2.14×10^{-8}	4.124
0.25	3.04×10^{-3}	1.001	1.23×10^{-9}	4.061
0.125	1.52×10^{-3}	1.001	7.34×10^{-11}	4.031
0.0625	7.59×10^{-4}	1.000	4.49×10^{-12}	4.015
0.03125	3.80×10^{-4}	–	2.78×10^{-13}	–
fitted slope		**1.001**		**4.085**

- Euler: error **halves** when h halves
- RK4: error falls by a factor of about **16**
- RK4's order sits slightly above 4 for large steps and approaches 4 as h shrinks

Result: observed orders 1.001 and 4.085 match the theory (1 and 4). At $h = 1$ day RK4 is about 3×10^4 times more accurate than Euler for four times the work per step, so RK4 is used for forward simulation.

PINN design: architecture and inductive biases

Module 4 ▪ two networks share one physics loss ▪ six design choices enforce SEIRD structure exactly

The observability problem

- Surveillance reports two series: cumulative cases C and deaths D
- Hidden compartments S, E, I are **never measured**
- $\beta(t)$ and μ must be inferred from C and D alone

Observable re-parameterisation

Set $C = I + R + D$ (everything symptomatic). Then $S = N - E - C$ exactly, so mass conservation *cannot be violated* for any network output.

Dual-network architecture

- **State net** $g_\theta(\tau)$: 3×48 tanh units, 4 outputs

$$e = e_0\, e^{\tau g_1},\ i = i_0\, e^{\tau g_2},\ c = c_0\, e^{\tau g_3},\ d = d_0 + \tau g_4$$

- **Transmission net** $h_\phi(\tau)$: 3×32 tanh units, $\beta(\tau) = \text{Softplus}(h_\phi(\tau) + b) > 0$

Six inductive biases

1. **Hard ICs:** exponential form $e_0 e^{\tau g_1}$ satisfies $e(0) = e_0$ exactly; τ factor means the gradient of the IC loss is always zero
2. **Positivity:** Softplus and exponential forms keep every state > 0
3. **CFR bounds:** $\mu = \gamma\,\text{CFR}/(1 - \text{CFR})$, $\text{CFR} \in (0.2, 0.8)$, blocks the degenerate $\mu \to 0$ solution
4. **Fourier features:** $\sin(k\pi\tau')$ and $\cos(k\pi\tau')$, $k = 1, \ldots, 4$ overcome spectral bias so $\beta(t)$ changes can be represented inside the window
5. **Relative ODE residuals:** each equation divided by its state, so the early (small) phase costs as much as the late (large) phase
6. **Smooth prior on** β**:** penalty on $(d\beta/d\tau)^2$ makes the inverse problem well-posed

What the architecture guarantees without training: $S + E + I + R + D = N$ to machine precision; $e, i, c > 0$; initial conditions exact; $\text{CFR} \in (0.2, 0.8)$.

Loss function and training protocol

Module 4 ▪ four terms; Adam brings it close, L-BFGS closes the gap

Total loss

$$\mathcal{L} = \underbrace{\mathcal{L}_C + \mathcal{L}_D}_{\text{data}} + \mathcal{L}_{\text{ode}} + w_{\text{sm}}\,\mathcal{L}_{\text{smooth}}$$

$$\mathcal{L}_C = \tfrac{1}{n_C}\textstyle\sum_k \big(c(\tau_k) - C_k/K\big)^2$$

$$\mathcal{L}_D = \tfrac{1}{n_D}\textstyle\sum_k \big(d(\tau_k) - D_k/K\big)^2$$

$$\mathcal{L}_{\text{ode}} = \tfrac{1}{N_f}\textstyle\sum_j \Big[\Big(\frac{e' - Tf_e}{e}\Big)^2 + \Big(\frac{i' - Tf_i}{i}\Big)^2 + \cdots\Big]_{\tau_j}$$

$$\mathcal{L}_{\text{smooth}} = \tfrac{1}{N_f}\textstyle\sum_j \big(\frac{d\beta}{d\tau}(\tau_j)\big)^2$$

K = max training case count; $N_f = 200$ collocation points; $w_{\text{sm}} = 10^{-4}$ (synthetic), 10^{-3} (real data); $w_{\text{data}} = w_{\text{ode}} = 1$.

Why relative ODE residuals?

An epidemic grows by orders of magnitude. With plain (absolute) residuals the early phase costs almost nothing. The E6 ablation shows absolute residuals give 75–128% forward error vs. below 2×10^{-3} with relative residuals.

Two-stage optimiser

- **Stage 1 – Adam** (LR 2×10^{-3}, cosine decay, 2000 iters): fresh random collocation points each iteration; fast escape from saddle points
- **Stage 2 – L-BFGS** (strong-Wolfe line search, 2500–6000 iters, fixed linspace grid): second-order convergence to a tight solution
- All weights in **double precision** (float64) for accurate physics residuals
- **Ensemble:** 5 independent seeds; median is the point estimate

Forecasting

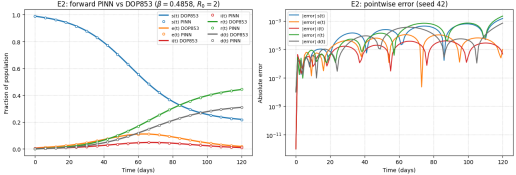
Hold β at its mean over the last 14 training days; keep learned μ; integrate forward with DOP853. Forecast is the median over seeds. The assumption that β stays constant limits predictive horizon.

E6 ablation headline

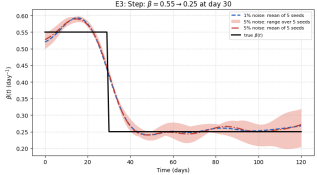
Adam-only: $e(t)$ error 96%; $w_{\text{ode}} = 10$: errors triple; no Fourier features: β RMSE +38%.

Synthetic experiments: E2 forward, E3 inverse $\beta(t)$, E4 hidden states

Module 4 ▪ no real data; ground truth is known; PINN is tested against DOP853



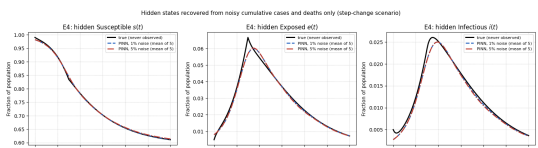
E2: forward PINN (markers) vs. DOP853 (lines) and pointwise error



E3: recovered $\beta(t)$ (band = seed range, 5% noise)

Quantity	1% noise	5% noise
$\beta(t)$ rel. L_2	$11.0\% \pm 0.7\%$	$11.6\% \pm 0.6\%$
μ rel. error	$4.6\% \pm 0.3\%$	$5.4\% \pm 1.0\%$
hidden $s(t)$	$0.36\% \pm 0.03\%$	$0.46\% \pm 0.09\%$
hidden $e(t)$	$5.8\% \pm 0.6\%$	$6.4\% \pm 0.7\%$
hidden $i(t)$	$5.8\% \pm 0.4\%$	$6.5\% \pm 0.8\%$

E3/E4: mean $\pm$ SD over 5 seeds; β drops 0.55$\to$0.25 at day 30

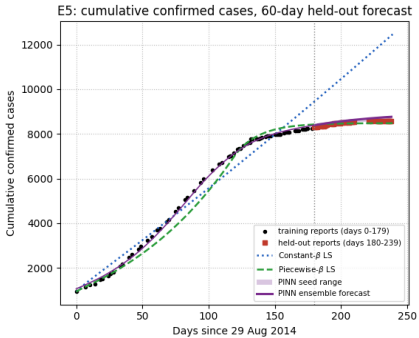


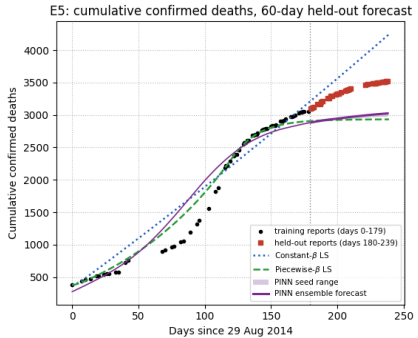
E4: reconstructed $s(t)$, $e(t)$, $i(t)$ from cases and deaths only

E2: worst compartment $< 2 \times 10^{-3}$; conservation 2×10^{-16} (machine zero). **E3:** a sharp β change recovered within 11%; 5$\times$ more noise costs $<$ 1 pp. **E4:** susceptible fraction recovered within 0.5% from *zero* direct measurements.

E5: Held-out forecast on WHO data

Module 5 ▪ WHO lab-confirmed reports, Sierra Leone
fit: 29 Aug 2014 – 24 Feb 2015 ▪ test: 25 Feb – 25 Apr 2015, never seen





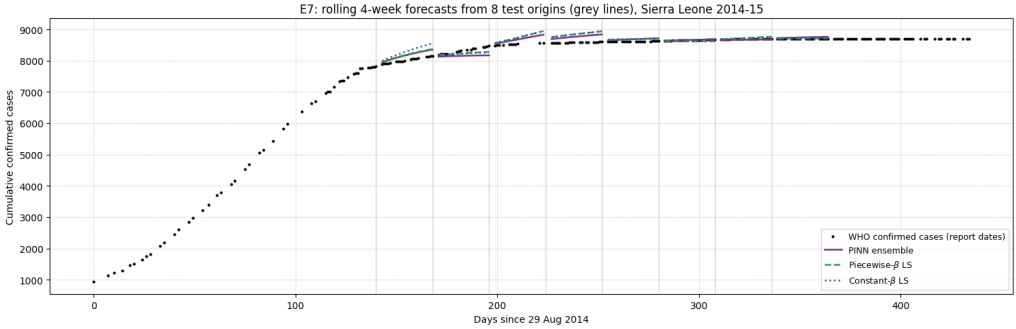
Cases, relative error	PINN	Piecewise-β	Constant-β
60-day forecast	1.35%	**0.82%**	28.6%

Deaths, relative error	PINN	Piecewise-β	Constant-β
60-day forecast	11.8%	12.9%	**10.7%**

A time-varying β is what matters: Constant-β keeps growing and misses cases by 29%. **PINN** is a close second to Piecewise-β on cases (single seeds: 0.28–1.75%). **Deaths defeat every method** (11–13%): WHO added 463 deaths but only 361 cases in the test window.

E7: Rolling 4-week forecasts across the epidemic

Module 5 ▪ 8 origins, every 28 days from 16 Jan to 31 Jul 2015 ▪ each fit sees only the previous 8 weeks



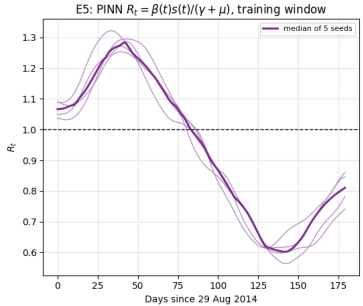
Mean of 8 origins	Cases MAE	New cases, 4 wk	Deaths MAE
PINN (3 seeds)	**96.8**	**148.2**	69.7
Piecewise-β	106.5	167.8	**42.0**
Constant-β	120.9	192.5	50.7

Errors in people. "New cases, 4 wk": error in new cases over the horizon.

✓ **Cases:** **PINN** is best on average, 20% below **Constant-β**, and beats **Piecewise-β** at 6 of 8 origins.

× **Deaths:** **PINN** is worst; its death forecast inherits any error in the learned μ.

What the PINN adds, and what it costs



Learned $R_t = \beta(t)\, s(t)/(\gamma+\mu)$, E5 training window (5 seeds, median bold):

- peak ≈ 1.28 in early Oct 2014
- crosses 1 in mid-Nov 2014: the epidemic turns
- ≈ 0.6 in mid-Jan 2015, ≈ 0.8 by late Feb 2015

What it gives that least squares does not

- a continuous $R_t(t)$ with no breakpoints to choose
- the case-fatality ratio, learned: 0.347–0.357 across seeds
- the hidden compartments S, E, I alongside the fit

What it costs

- 1–2 min per fit (Colab CPU) vs. seconds for least squares
- needs a seed ensemble: single seeds vary (0.28–1.75%)
- forecasts hold β at its recent value, so they cannot foresee a new intervention or a resurgence

Honest verdict: on cases the PINN forecasts about as well as the best classical fit; on deaths it does worse. Its real value is an interpretable, continuous picture of transmission.